

Data Visualization & Insight Communication



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With assistance from AI tools, particularly
Microsoft Copilot and Google Gemini



Outline

1

Eight analytical purposes

Trend, comparison, ranking, part-to-whole, distribution, relationship, geography, target.

2

Insight communication

Question → Visualization
→ Observation → Insight
→ Implication.

3

Visualization principles

Effective Data
Visualization Principles.

Pattern: question → chart choice → visualization → interpretation → best practice → code.

Why visualization is essential to analysis

Three Core Reasons:

- **It's Faster:** Visuals reveal patterns, such as trends, clusters, and outliers, much faster than tables of numbers.
- **It Reveals the Data:** Summary statistics can hide important differences. A plot shows the actual shape and patterns in the data.
- **It Supports Thinking:** Plotting data helps you explore, question, and discover insights - not just present results.

Motto: Visualize Before Analyzing!

- Charts reveal what summary statistics hide.
- Spot patterns, distributions, outliers, and relationships fast.
- [Anscombe's quartet](#): four datasets with identical mean, SD, and regression line - yet completely different shapes.
 - Always plot the data before you trust the statistics.



Eight Analytical Purposes

For each purpose:

question → chart options → example → how to read it → best practice & common mistakes → code



The eight purposes at a glance

1

Trend

How does it change over time?

Line chart

2

Comparison

How do categories differ?

Bar chart

3

Ranking

What is highest or lowest?

Sorted bar

4

Part-to-Whole

How does each part contribute?

Stacked bar

5

Distribution

How are values spread?

Histogram / box

6

Relationship

How are variables related?

Scatter plot

7

Geographic

Where does it happen?

Choropleth map

8

Target / Deviation

How do we compare with plan?

Actual-vs-target bar



Purpose 1 - Trend

Analytical Question

How does the metric change over time?

Recommended charts - and when to use each

★ DEFAULT - LINE CHART

Shows slope, turning points and seasonality.
Use for 5+ time points.

ALTERNATIVE - AREA CHART

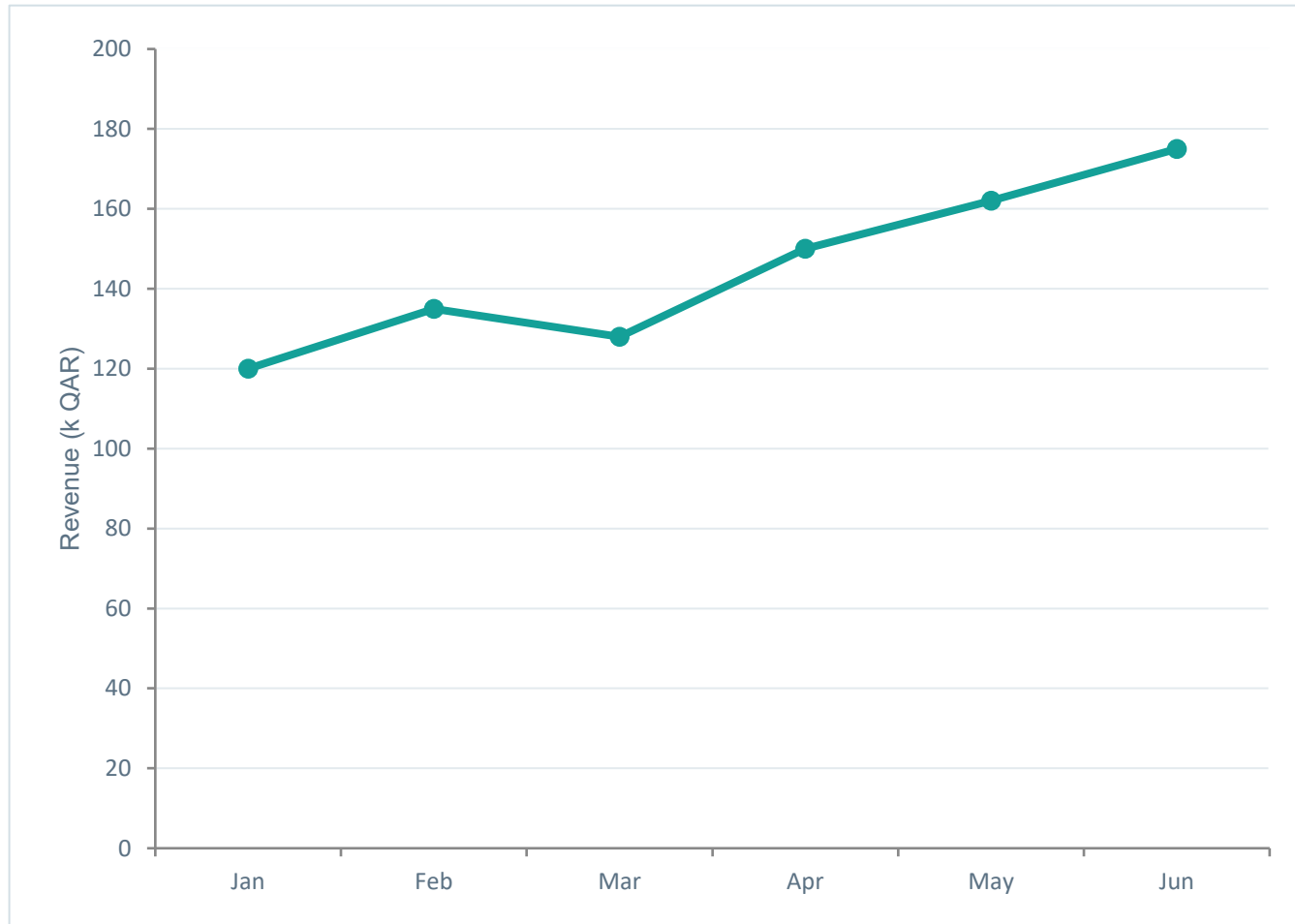
Use when the size of a total matters. Avoid overlapping areas.

ALTERNATIVE - COLUMN CHART

Use when periods are few and discrete, e.g. 4 quarters.



Trend - example and interpretation



ShopStream monthly revenue, Jan–Jun

How to read it

Steeper **slope** means faster change.
Check the y-axis starts at zero.

What to look for

Direction, rate of change, turning points, seasonality, and any flat or missing segment.

✓ Justified conclusion

"Revenue rose 46% over six months, with the fastest growth between April and June."

✗ Not justified

"Revenue will reach 250k by December." Six points cannot support an extrapolation, and no seasonal cycle has been observed yet.



Trend - best practice and common mistakes

✓ Best Practices

- Keep time on the x-axis, evenly spaced, oldest on the left.
- Start the value axis at zero for amounts.
- Limit to 3–5 lines; highlight one and mute the rest.
- Label the final point directly instead of using a legend.
- State the aggregation level in the title: daily, weekly, monthly.

✗ Common mistakes

- Truncated y-axis that turns a 3% change into a cliff.
- Uneven time intervals plotted as if evenly spaced.
- Connecting points across a gap where data is missing.
- Spaghetti charts: twelve lines, no hierarchy.



Trend - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'revenue': [120, 135, 128, 150, 162, 175]})

sns.set_theme(style='whitegrid')
ax = sns.lineplot(data=df, x='month', y='revenue',
                  marker='o', linewidth=2.5)
ax.set_ylim(0, 200)
ax.set(xlabel='', ylabel='Revenue (k QAR)')
ax.set_title('Revenue grew 46% in H1 2025')
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'revenue': [120, 135, 128, 150, 162, 175]})

fig = px.line(df, x='month', y='revenue',
               markers=True, title='Revenue grew 46% in H1 2025')
fig.update_yaxes(range=[0, 200],
                  title='Revenue (k QAR)')
fig.update_xaxes(title='')
fig.show()
```

Seaborn returns a static Matplotlib chart → ideal for reports and PDFs. Plotly returns an interactive figure with hover values → ideal for dashboards.



Purpose 2 - Comparison

Analytical Question

How do categories differ from one another?

Recommended charts - and when to use each

★ DEFAULT - BAR / COLUMN CHART

Uses bar length from a **zero baseline** to represent values, making differences easy and accurate to compare.

ALTERNATIVE - GROUPED BAR CHART

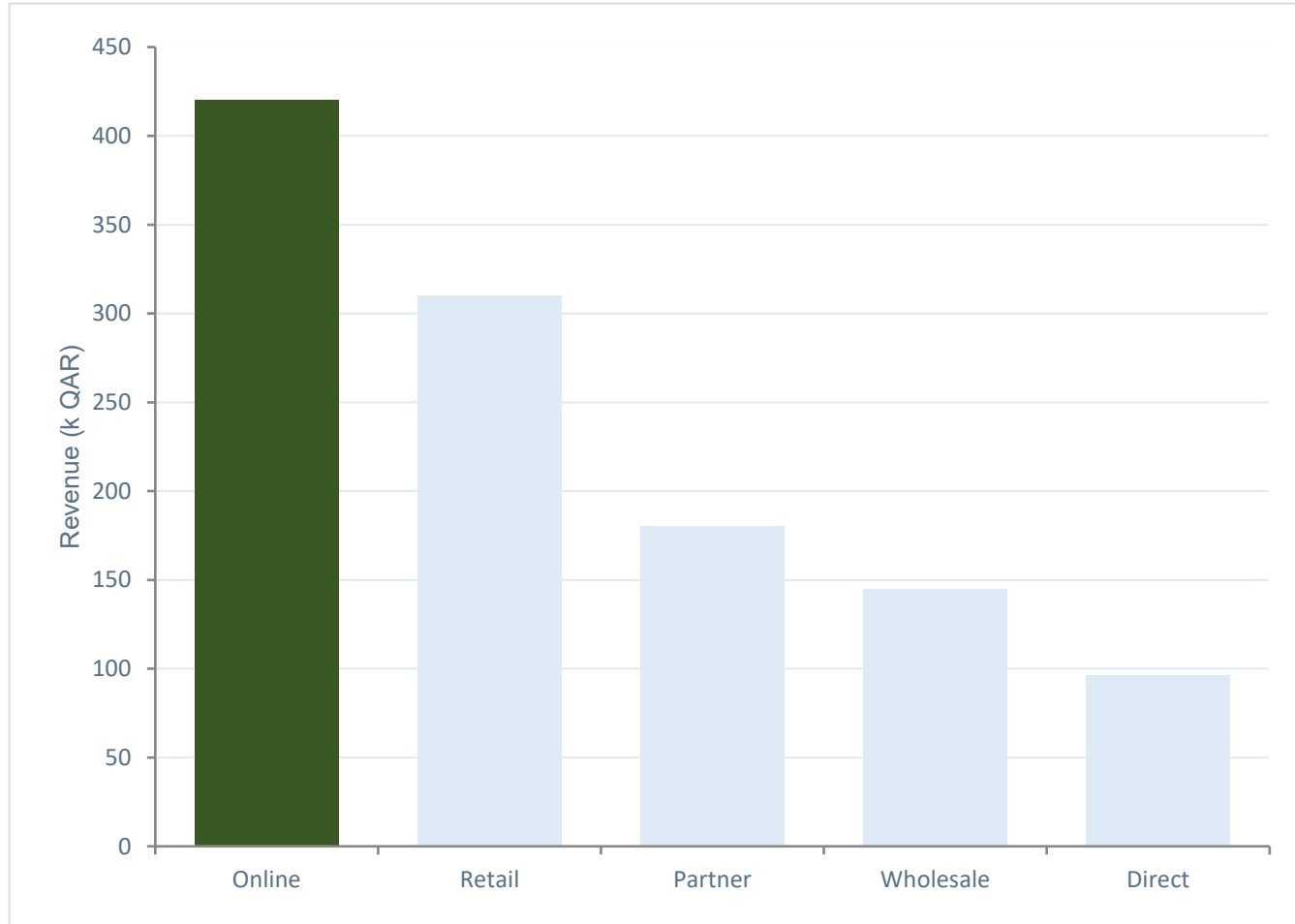
Use when a category splits by a second variable. Keep 2–3 groups.

ALTERNATIVE - DOT PLOT

Use for many categories or long labels.



Comparison - example and interpretation



Revenue by sales channel, H1

How to read it

Compare lengths on a shared zero baseline: length is proportional to value.

What to look for

Which categories lead, the size of the gap, and whether a few categories dominate the total.

✓ Justified conclusion

"Online generates 4.4× the revenue of Direct and more than a third of H1 revenue."

✗ Not justified

"Online is our most profitable channel." Revenue is not profit; cost of acquisition is not on this chart.



Comparison - best practice and common mistakes

✓ Best Practices

- Always start the value axis at zero.
- Order bars by value unless the categories have a natural order.
- Use horizontal bars when labels are long.
- One color for all bars; a second color only to highlight the point.

✗ Common mistakes

- Truncated axis exaggerates differences by hiding the zero baseline.
- Rainbow coloring that imply differences or meaning that the colors do not actually represent.
- 3D bars, which make the front row look larger.
- Alphabetical Sorting hides magnitude differences when the goal is comparison.

Comparison - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'channel': ['Online', 'Retail', 'Partner',
               'Wholesale', 'Direct'],
    'revenue': [420, 310, 180, 145, 96]})

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, x='channel',
                 y='revenue', color='#14A098')
ax.set(xlabel='', ylabel='Revenue (k QAR)')
ax.set_title('Online delivers 4.4x the revenue of
Direct')

plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'channel': ['Online', 'Retail', 'Partner',
               'Wholesale', 'Direct'],
    'revenue': [420, 310, 180, 145, 96]})

fig = px.bar(df, x='channel', y='revenue',
             text_auto=True,
             title='Online delivers 4.4x the
revenue of Direct')

fig.update_traces(marker_color='#14A098')
fig.update_layout(yaxis_title='Revenue (k QAR)',
                 xaxis_title='')

fig.show()
```

Note the single color in both libraries. Default palettes assign a different color per bar.



Purpose 3 - Ranking

Analytical Question

What is highest, what is lowest, and in what order?

Recommended charts - and when to use each

★ DEFAULT - SORTED HORIZONTAL BAR

Sorting does the analytical work
Horizontal fits long names.

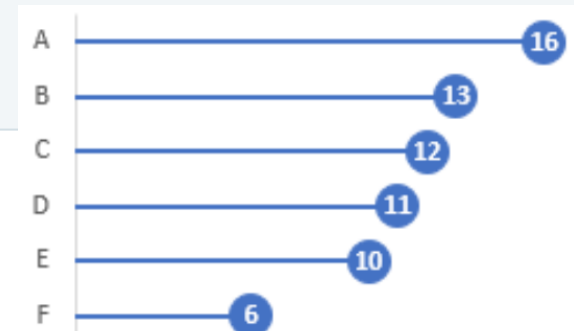
ALTERNATIVE - TOP-N CHART

Use when only leaders matter. Always add an 'Other' bar.



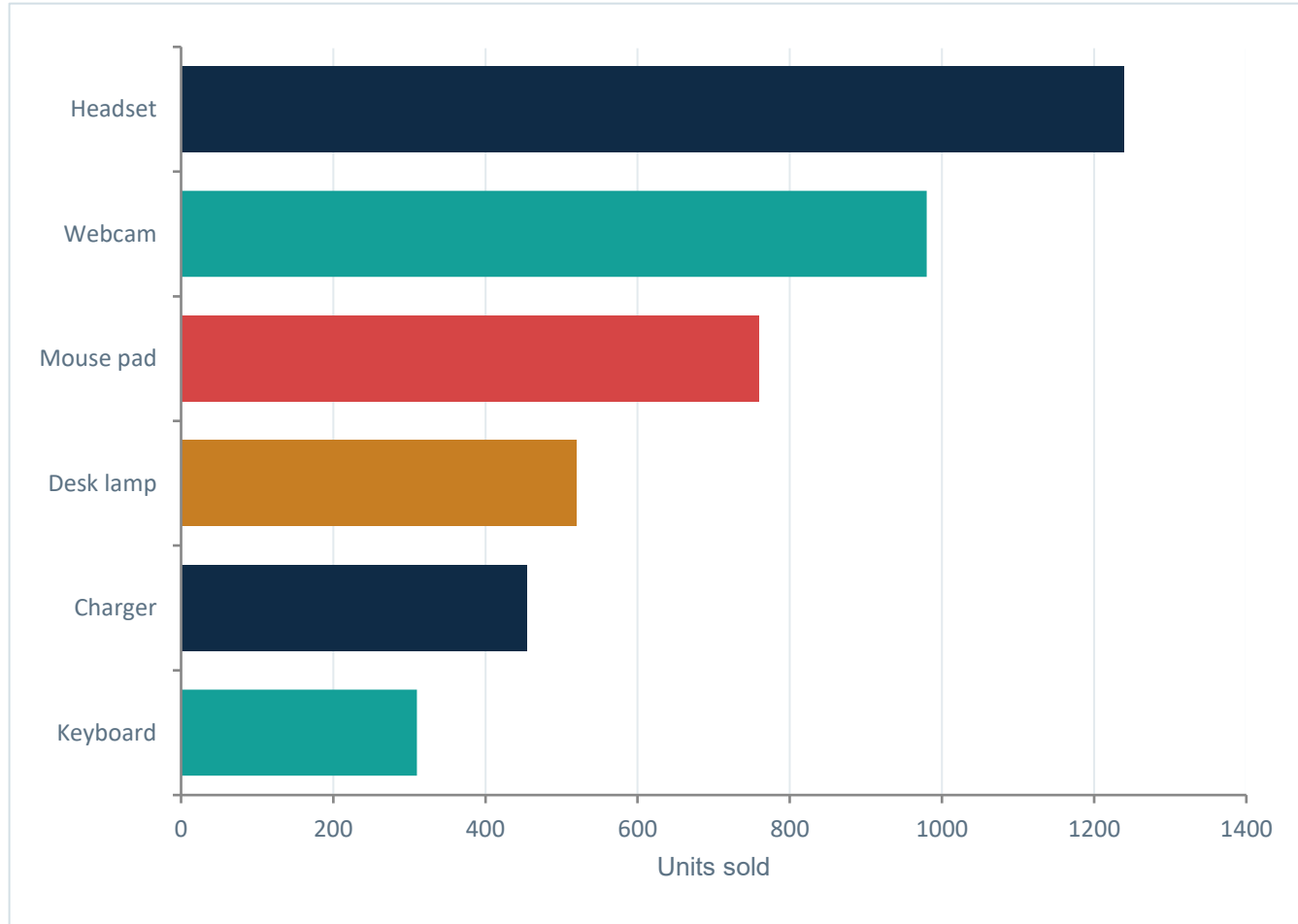
ALTERNATIVE - LOLLIPOP CHART

Same information, less ink, for 15–30 items.





Ranking - example and interpretation



Top six products by units sold, Q2 (sorted)

How to read it

Read from the longest bar down. The order is the message; gaps show how big the lead is.

What to look for

The leader, the size of the gap to second place, and where the curve flattens into a long tail.

✓ Justified conclusion

"Headsets lead on volume at 1,240 units, 27% ahead of webcams; the top three account for most unit sales."

✗ Not justified

"Headsets are our best product." Units are not revenue or margin - a cheap high-volume item can be the least valuable.



Ranking - best practice and common mistakes

✓ Best Practices

- Sort descending; never leave the default alphabetical order.
- Use horizontal bars so names read naturally.
- Show the full ranking or state explicitly how many were cut.
- Highlight only the item the narrative is about.
- Add a value label when exact figures matter to the decision.

✗ Common mistakes

- Unsorted bars - the reader has to rank them by eye.
- Hidden Top-N: showing five of forty with no 'Other' bar.
- Ranking by a volatile metric on a tiny sample, so the order changes weekly.
- Treating a 2% gap between ranks 1 and 2 as a meaningful difference.
- Ranking on a metric that is not the one the decision depends on.



Ranking - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'product': ['Headset', 'Webcam', 'Mouse pad',
               'Desk lamp', 'Charger', 'Keyboard'],
    'units': [1240, 980, 760, 520, 455, 310]})
df = df.sort_values('units', ascending=False)

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, y='product', x='units',
                 orient="h", color='#C77E23')
ax.set(ylabel='', xlabel='Units sold')
ax.set_title('Headsets lead Q2 volume, 27% ahead of webcams')
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'product': ['Headset', 'Webcam', 'Mouse pad',
               'Desk lamp', 'Charger', 'Keyboard'],
    'units': [1240, 980, 760, 520, 455, 310]})

fig = px.bar(df, x='units', y='product',
             orientation='h', title='Headsets lead Q2 volume')
fig.update_yaxes(categoryorder='total ascending')
fig.update_traces(marker_color='#C77E23')
fig.show()
```

Sort: Pandas **sort_values** in Seaborn, versus Plotly's **categoryorder='total ascending'**.
Use 'ascending' for horizontal bars because Plotly draws the first category at the bottom.

Purpose 4 - Part-to-Whole

Analytical Question

How does each part contribute to the total?

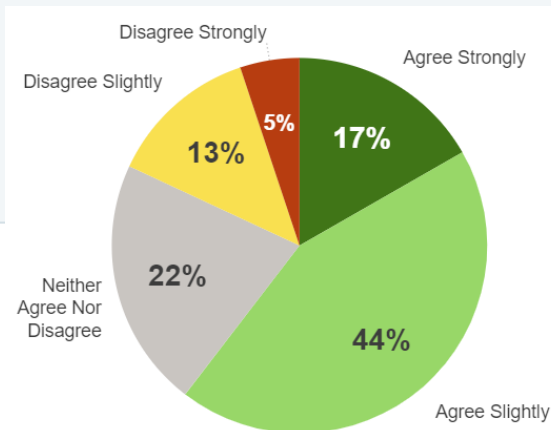
Recommended charts - and when to use each

★ DEFAULT - 100% STACKED BAR

Shows how each category contributes to a whole, with every bar totaling 100%.

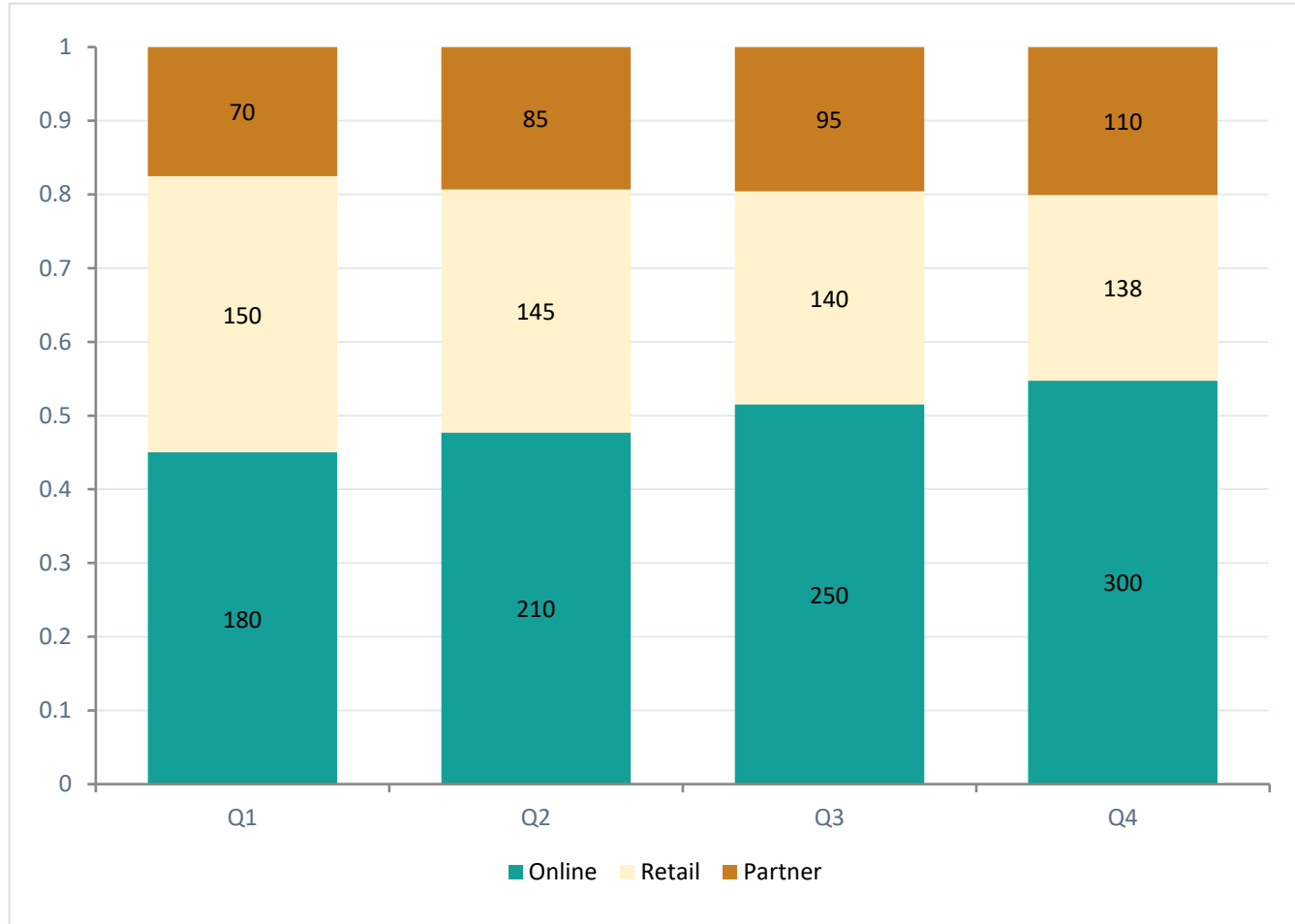
ALTERNATIVE - PIE / DONUT

Only with 2–4 parts and large differences. Angle and area are read poorly; label every slice with its share.





Part-to-Whole - example and interpretation



Share of revenue by channel, 100% stacked by quarter

How to read it

Each bar totals 100%. Read one band across bars. Only the bottom band has a stable baseline.

What to look for

Which share grows, which shrinks, and whether the mix is shifting steadily or abruptly.

✓ Justified conclusion

"Online's share of revenue rose from 45% in Q1 to 55% in Q4, while Retail's share fell by roughly ten points."

✗ Not justified

"Retail is shrinking." Retail's **share** fell; in absolute terms it was broadly flat. Share and amount are different claims.

Part-to-Whole - best practice and common mistakes

✓ **Best Practices**

- Only use part-to-whole when the parts genuinely sum to a meaningful total.
- Put the series you most want tracked at the bottom, on the stable baseline.
- Limit to 3–5 segments; group the rest into 'Other'.
- Label shares directly as percentages.
- Show the absolute total somewhere if the decision depends on size, not just mix.

✗ **Common mistakes**

- Pies with ten slices and a legend the reader must decode.
- Stacking categories that overlap or do not sum to the whole.
- Two pies side by side to compare change - use a stacked bar instead.
- Exploded or 3D pies, which distort the area badly.
- Confusing a falling share with a falling amount.



Part-to-Whole - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, matplotlib.pyplot as plt,
seaborn as sns

df = pd.DataFrame({
    'quarter': ['Q1', 'Q2', 'Q3', 'Q4'],
    'Online': [180, 210, 250, 300],
    'Retail': [150, 145, 140, 138],
    'Partner': [70, 85, 95, 110]})
df.set_index('quarter')

# Convert revenue to percentage share within each quarter
share = df.div(df.sum(axis=1), axis=0) * 100
sns.set_theme(style='whitegrid')
ax = share.plot(kind='bar', stacked=True,
                figsize=(7,4))
ax.set(xlabel='', ylabel='Share of revenue (%)')
ax.set_title("Online's share rose from 45% to 55%")
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'quarter': ['Q1', 'Q2', 'Q3', 'Q4']*3,
    'channel': ['Online']*4 + ['Retail']*4 +
    ['Partner']*4,
    'revenue': [180,210,250,300, 150,145,140,138,
                70,85,95,110]})

fig = px.bar(df, x='quarter', y='revenue',
             color='channel', barnorm='percent',
             title="Online's share rose from 45% to 55%")
fig.update_layout(yaxis_title='Share of revenue (%)')
fig.show()
```

Seaborn has no stacked-bar API, so we use Pandas' **.plot** on a wide frame.

Plotly takes long (tidy) format and normalises with **barnorm='percent'**.



Purpose 5 - Distribution

Analytical Question

How are the values of one variable spread out?

Recommended charts - and when to use each

★ DEFAULT - HISTOGRAM

The default for one variable. Shows centre, spread, skew and gaps. The bin width is an analytical choice.

ALTERNATIVE - BOX PLOT

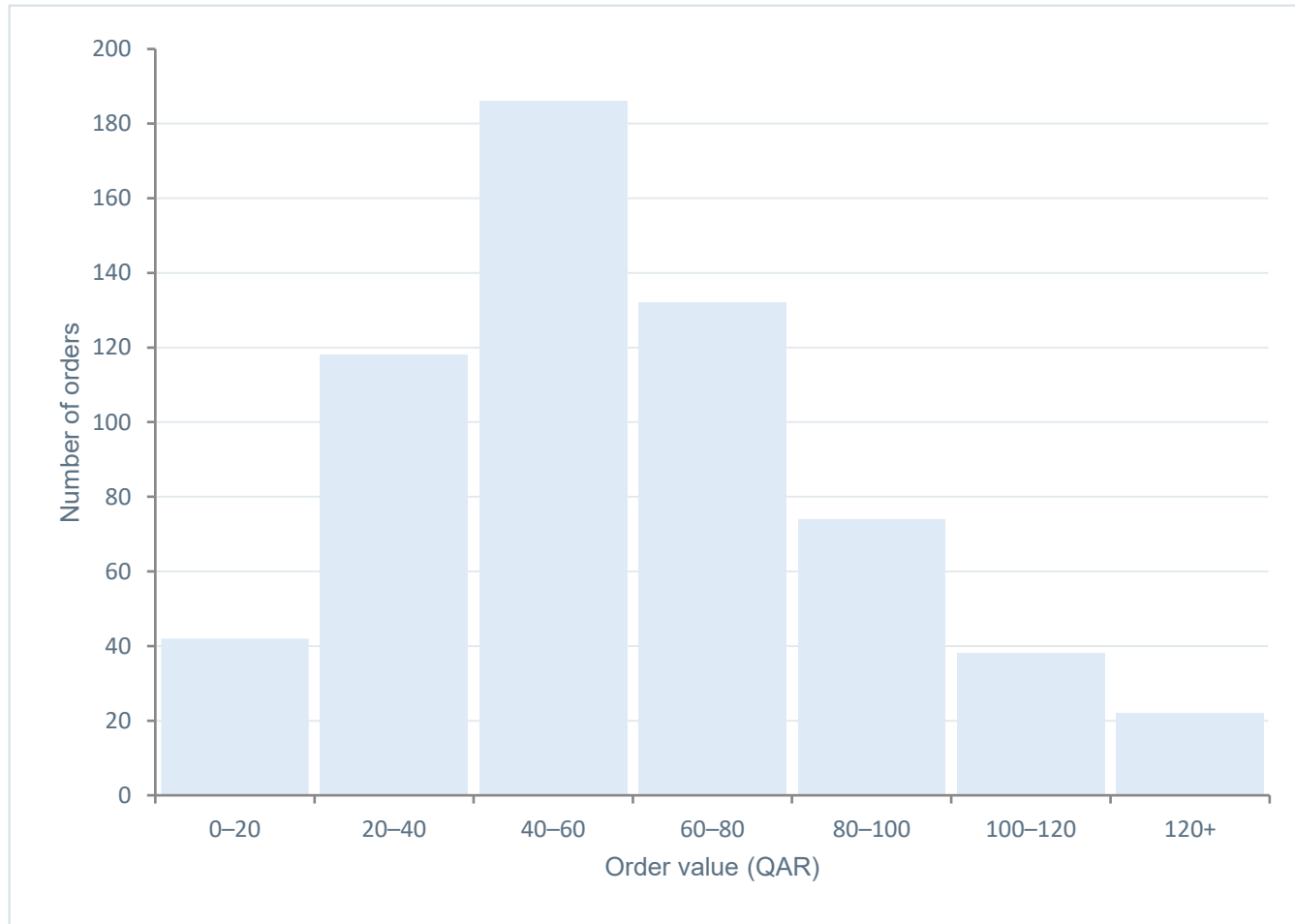
Best for comparing distributions across groups compactly. Hides multimodality.

ALTERNATIVE - VIOLIN / STRIP PLOT

When shape matters or the sample is small enough to show individual points.



Distribution - example and interpretation



Distribution of order values, 612 orders

How to read it

Height is frequency, not amount. The x-axis is a measurement cut into bins.

What to look for

Centre, spread, skew direction, number of peaks, and a long tail.

✓ Justified conclusion

"Order values are right-skewed: most orders fall between 20 and 80 QAR, with a thin tail of high-value orders above 120."

✗ Not justified

"The average order is 60 QAR, so a typical customer spends 60." With right skew, the mean is pulled up by the tail - report the median.

Distribution - best practice and common mistakes

✓ Best Practices

- Try several bin widths; too few hides structure, too many shows noise.
- Label the axes so frequency is never mistaken for amount.
- Report the median and IQR alongside a skewed distribution.
- Use a box plot when comparing many groups, a histogram for one.
- Say how many observations the distribution is based on.

✗ Common mistakes

- Reporting only a mean for a skewed variable.
- Cutting off the tail and hiding the outliers that matter most.
- Comparing histograms with different bin widths or sample sizes.

Distribution - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, numpy as np, seaborn as sns
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
orders = pd.DataFrame({
    'value': rng.gamma(shape=3, scale=18,
size=612)})

sns.set_theme(style='whitegrid')
ax = sns.histplot(data=orders, x='value', bins=20,
                  color='#14A098')
ax.axvline(orders['value'].median(),
          color='#D64545', linestyle='--', label='Median')
ax.set(xlabel='Order value (QAR)',
      ylabel='Number of orders')
ax.legend(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd, numpy as np
import plotly.express as px

rng = np.random.default_rng(42)
orders = pd.DataFrame({
    'value': rng.gamma(shape=3, scale=18,
size=612)})

fig = px.histogram(orders, x='value', nbins=20,
                  marginal='box',
                  title='Order values are right-skewed')
fig.update_layout(xaxis_title='Order value (QAR)',
                  yaxis_title='Number of orders')
fig.show()
```

Plotly's `marginal='box'` stacks a box plot above the histogram, allowing to see both views of the same variable at once.

Purpose 6 - Relationship

Analytical Question

How are two (or more) variables related?

Recommended charts - and when to use each

★ DEFAULT - SCATTER PLOT

The default for two numeric variables. Shows direction, strength, form and outliers at once.

ALTERNATIVE - BUBBLE CHART

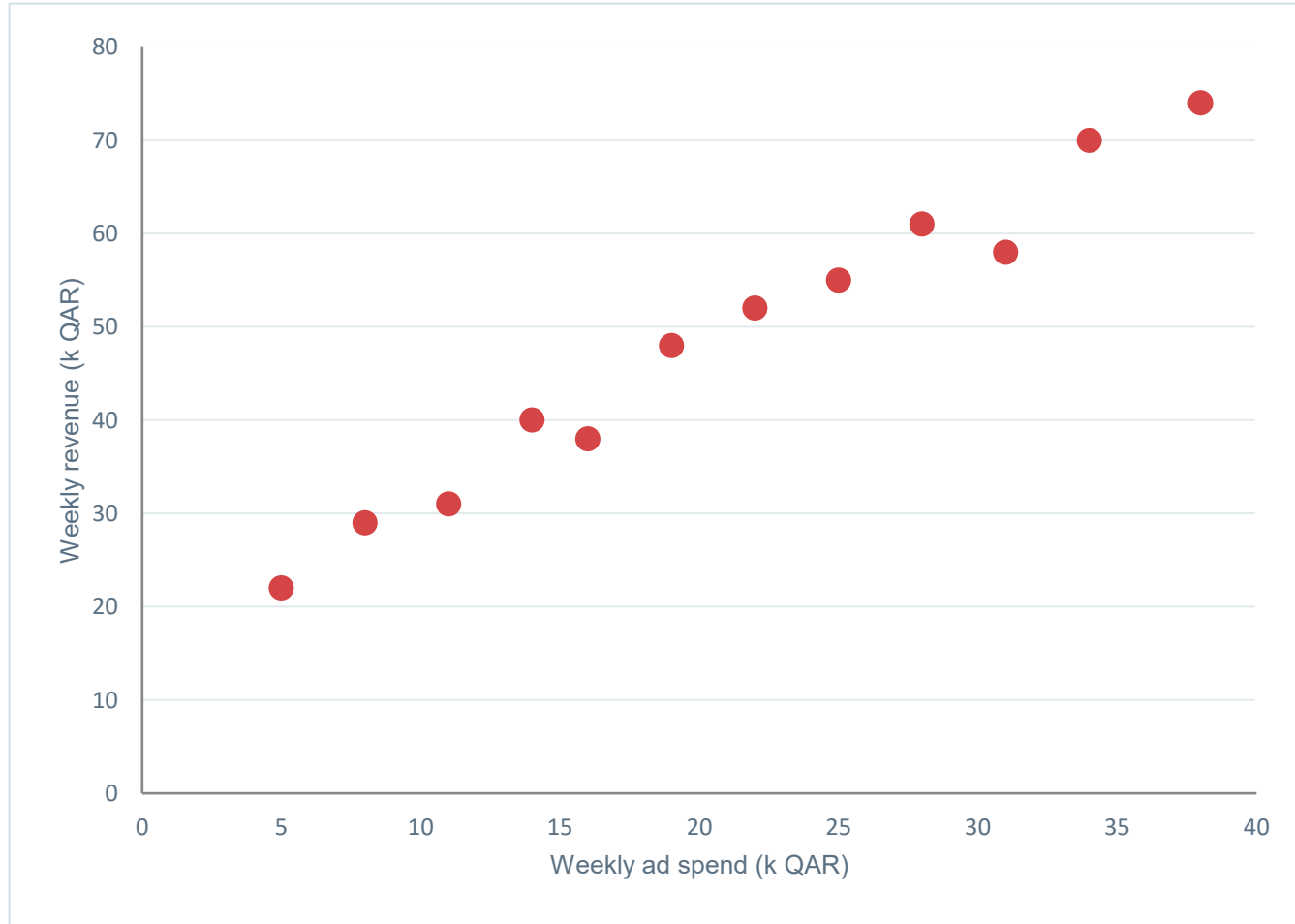
Adds a third variable as point size - use sparingly; area is read inaccurately.

ALTERNATIVE - HEATMAP

For a correlation matrix across many variables, as a screening tool before detailed plots.



Relationship - example and interpretation



Weekly ad spend vs revenue, 12 weeks

How to read it

Each point is one week: x = spend, y = revenue. Read the cloud's shape, not single points.

What to look for

Direction (positive/negative), strength (tight or loose), form (straight or curved), and points far from the pattern.

✓ Justified conclusion

"Weeks with higher ad spend tend to show higher revenue; the relationship is positive and fairly strong over this range."

✗ Not justified

"Ad spend causes revenue to rise, so doubling spend doubles revenue." Correlation is not causation, and extrapolating beyond 38k is unsupported.

Relationship - best practice and common mistakes

✓ Best Practices

- Plot the data before computing a correlation coefficient.
- Put the presumed driver on the x-axis.
- Add a trend line only if the form genuinely looks linear.
- State the unit of observation: a week, a customer, a store?

✗ Common mistakes

- Claiming Causation: A correlation does not prove that one variable causes the other.
- Reporting r without scatter plot: $r = 0.8$ may hide a curved pattern or be driven by an outlier.
- Ignoring Lurking Variables: A third variable may explain the relationship - for example, both sales and ads rise during holidays.
- Extrapolating: Do not extend a fitted line beyond the observed data range.

Relationship - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'ad_spend': [5,8,11,14,16,19,22,25,28,31,34,38],
    'revenue': [22,29,31,40,38,48,52,55,61,58,70,74]})

sns.set_theme(style='whitegrid')
ax = sns.regplot(data=df, x='ad_spend',
    y='revenue', color='#D64545', ci=None)
ax.set(xlabel='Weekly ad spend (k QAR)',
    ylabel='Weekly revenue (k QAR)')
ax.set_title('Higher ad spend is associated with
higher revenue')
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'ad_spend': [5,8,11,14,16,19,22,25,28,31,34,38],
    'revenue': [22,29,31,40,38,48,52,55,61,58,70,74]})

fig = px.scatter(df, x='ad_spend', y='revenue',
    trendline='ols',
    title='Ad spend and revenue move together')
fig.update_layout(
    xaxis_title='Weekly ad spend (k QAR)',
    yaxis_title='Weekly revenue (k QAR)')
fig.show()
```

Both add a fitted line in one argument - which is exactly why the wording of the title matters. 'Associated with' is defensible; 'causes' is not.



Purpose 7 - Geographic

Analytical Question

Where does the pattern happen, and does location explain it?

Recommended charts - and when to use each

★ DEFAULT - CHOROPLETH MAP

Regions shaded by value. Use rates per capita, never raw counts - otherwise you are mapping population.

ALTERNATIVE - SYMBOL / BUBBLE MAP

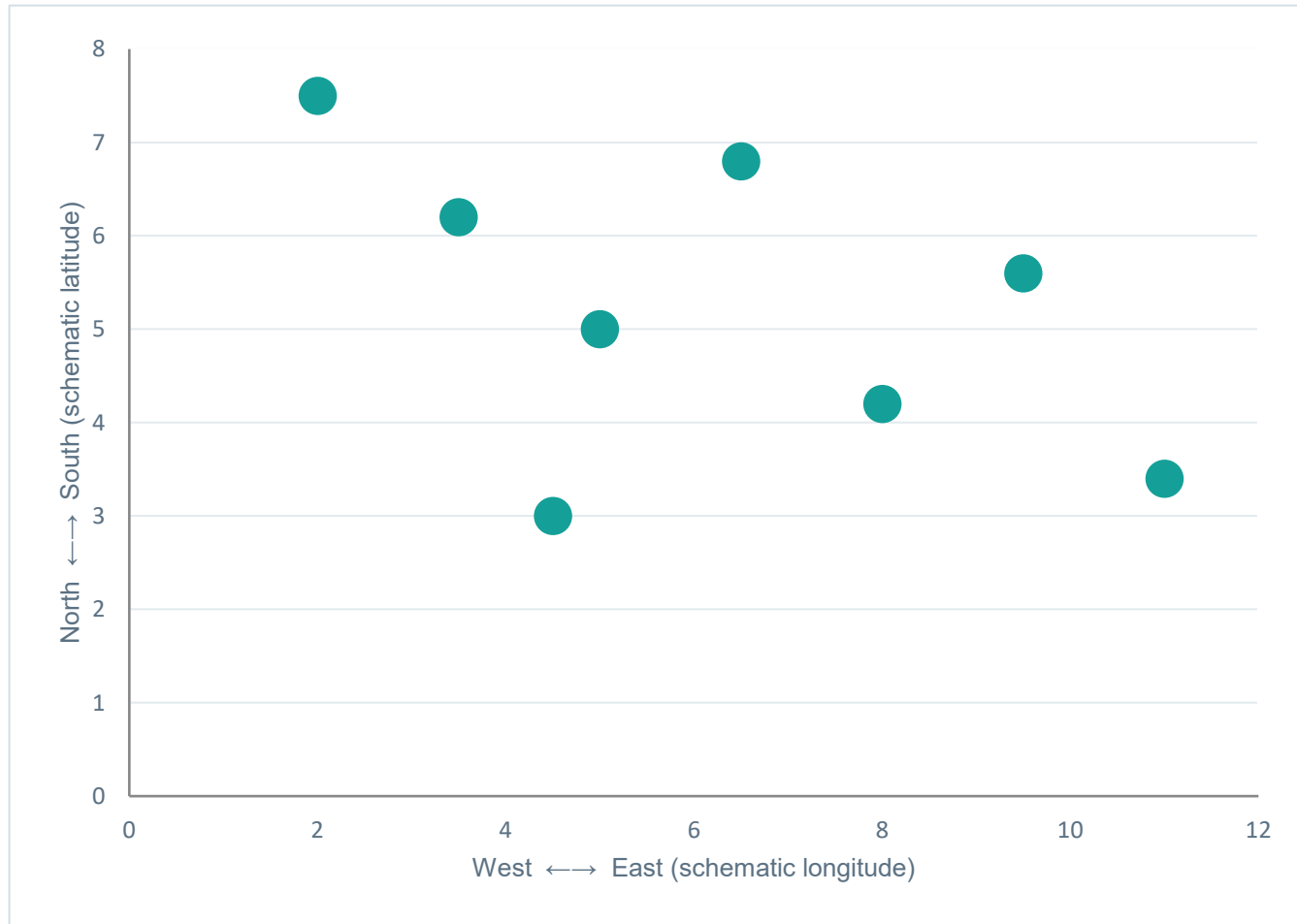
Points sized by value. Better for raw counts and for small areas that a choropleth would hide.

ALTERNATIVE - SORTED BAR BY REGION

Often clearer than a map when you have fewer than ~15 regions and location itself is not the insight.



Geographic - example and interpretation



Schematic symbol map: eight city hubs by position

How to read it

Each marker is a city at its coordinates.
Position shows where, not how much.

What to look for

Clusters, isolated hotspots, and whether neighbouring regions behave alike (spatial autocorrelation).

✓ Justified conclusion

"Order rates per 1,000 residents are highest in the northern cluster of hubs, which sit closest to the distribution centre."

✗ Not justified

"The north buys more because northerners prefer our products." Proximity to the warehouse - faster delivery - is a far simpler explanation.



Geographic - best practice and common mistakes

✓ Best Practices

- Normalise to a rate: per capita, per household, per 1,000 customers.
- Use a sequential color scale for magnitude, diverging only around a true midpoint.
- State the geographic unit and the time window.
- Keep map projection and color bins consistent across comparisons.
- If the map adds nothing a bar chart could not, use the bar chart.

✗ Common mistakes

- Choropleths of raw counts - a population map in disguise.
- Large rural areas dominating visually while dense cities vanish.
- Arbitrary color bins chosen to manufacture a pattern.
- Ecological fallacy: inferring individual behaviour from area averages.
- Using a map for decoration when location is irrelevant to the question.

Geographic - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
# Seaborn has no map projection; use lat/lon as a
# schematic scatter, or GeoPandas for real geometry.
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'city':
    ['Doha', 'Dubai', 'Riyadh', 'Kuwait', 'Muscat'],
    'lon': [51.5, 55.3, 46.7, 47.9, 58.4],
    'lat': [25.3, 25.2, 24.7, 29.4, 23.6],
    'rate': [7.5, 6.2, 5.0, 6.8, 4.2]})

sns.set_theme(style='whitegrid')
ax = sns.scatterplot(data=df, x='lon', y='lat',
                    size='rate', hue='rate',
                    sizes=(60,400),
                    palette='viridis', legend='brief')
ax.set_title('Order rate per 1,000 residents by city')
plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'iso': ['QAT', 'ARE', 'SAU', 'KWT', 'OMN'],
    'rate': [7.5, 6.2, 5.0, 6.8, 4.2]})

fig = px.choropleth(df, locations='iso',
                    color='rate',
                    color_continuous_scale='Viridis',
                    scope='asia',
                    title='Order rate per 1,000
                    residents')
fig.update_layout(coloraxis_colorbar_title='per
                    1,000')
fig.show()
```

This is the one purpose where the libraries genuinely differ: Plotly draws real projected maps out of the box, while Seaborn needs GeoPandas. Choose the tool to fit the question.

Purpose 8 - Target / Deviation

Analytical Question

How does actual performance compare with a target or benchmark?

Recommended charts - and when to use each

★ DEFAULT - ACTUAL-VS-TARGET BAR

Bars for actual with a target marker or line. The clearest option for most audiences.

ALTERNATIVE - BULLET CHART

Compact: actual bar, target tick, qualitative bands. Ideal for dashboards with many KPIs.

ALTERNATIVE - VARIANCE CHART

Plots the gap itself - above and below zero - when the size of the miss is the message.

Target / Deviation - example and interpretation



Q4 revenue versus target by region

How to read it

Bars are actuals; the marker is the target.
The gap between them is the message.

What to look for

Who is above and below, the size of each gap, and whether misses cluster in one part of the business.

✓ Justified conclusion

"EMEA finished 12% below a 450k target, the only material miss; APAC beat target by 11%."

✗ Not justified

"EMEA underperformed, so the EMEA team is failing." A target may have been set unrealistically, or the market may have moved.

Target / Deviation - best practice and common mistakes

✓ Best Practices

- Show the target explicitly; a bar alone cannot answer the question.
- Use a consistent color rule: one color for above target, one for below.
- Label the variance as both an amount and a percentage.
- State where the target came from and when it was set.
- Order regions by variance so the biggest gap is impossible to miss.

✗ Common mistakes

- Plotting actuals with no target at all.
- Moving the target mid-period so every result looks like a success.
- Red/green as the only signal - invisible to many color-blind readers.
- Mixing percentage and absolute variance on one axis.
- Treating a 1% miss and a 30% miss as equally serious.



Target / Deviation - implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'region': ['North', 'South', 'EMEA', 'APAC', 'LATAM'],
    'actual': [520, 480, 396, 610, 340],
    'target': [500, 500, 450, 550, 350]})
df['miss'] = df['actual'] < df['target']

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, x='region', y='actual',
                 hue='miss',
                 palette=['#14A098', '#D64545'],
                 dodge=False)
ax.scatter(df['region'], df['target'], marker='_',
          s=900, color='black', label='Target')
ax.set(xlabel='', ylabel='Revenue (k QAR)')
ax.legend(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.graph_objects as go

df = pd.DataFrame({
    'region': ['North', 'South', 'EMEA', 'APAC', 'LATAM'],
    'actual': [520, 480, 396, 610, 340],
    'target': [500, 500, 450, 550, 350]})

fig = go.Figure()
fig.add_bar(x=df.region, y=df.actual, name='Actual',
            marker_color='#14A098')
fig.add_scatter(x=df.region, y=df.target, name='Target',
                mode='markers',
                marker=dict(symbol='line-ew', size=28,
                           line=dict(width=3,
                                       color='black'))))
fig.update_layout(yaxis_title='Revenue (k QAR)',
                  title='EMEA is the only material miss (-12%)')
fig.show()
```

Plotly Express cannot easily mix bars and markers, so we drop to graph_objects. Knowing when to leave the express API is a genuine skill.

Insight Communication

A chart is evidence. Explain what it shows, what it means, and what to do next.

A five-step structure that works every time

1. Question

What decision needs an answer?

Are we growing, and where?

2. Visualization

Which chart answers it?

Monthly revenue line chart

3. Observation

What does the chart show?

Revenue rose from 120k to 175k.

4. Insight

What does the pattern mean?

Growth accelerated after May, the metric increased faster each month.

5. Implication

Why does it matter, what next?

Plan Q3 capacity for the higher demand level, and verify whether the pricing change drove the acceleration.



Observation vs insight vs implication

OBSERVATION

What the chart shows

"Online revenue grew 46% between January and June, while Retail was flat."

Descriptive · verifiable from the chart alone · no domain knowledge required.

INSIGHT

What the pattern means

"Growth is coming entirely from one channel, so the business is becoming dependent on Online."

Interpretive · compares the pattern to an expectation · needs context.

IMPLICATION

Why it matters, and what to do

"Concentration risk is rising: protect Online margins and test a second growth channel this quarter."

Decision-oriented · names an action or a risk.

Test: identical for any dataset → not an insight. No consequence named → not an implication.

Before → after: turning a description into an insight

Order values

WEAK

"The histogram shows the distribution of order values across the dataset."

STRONG

"Order values are right-skewed: half of all orders are under 48 QAR, but a 6% tail above 120 QAR drives a quarter of revenue - so free-shipping thresholds should be set from the median, not the mean."

Regional targets

WEAK

"The bar chart compares actual and target revenue for five regions in Q4."

STRONG

"EMEA was the only material miss at 12% below target (-54k); the other four regions met or beat plan, so this is a regional issue, not a company-wide shortfall."

Ad spend

WEAK

"There is a correlation of 0.93 between ad spend and revenue."

STRONG

"Higher-spend weeks are consistently higher-revenue weeks across 5–38k of spend; the association is strong but untested as causal, so a split-region test should run before budget is increased."

Strong statements: a quantified finding + a stated limit + a consequence.



Chart-selection guide: purpose → visualization

Purpose	Analytical question	Recommended	Use instead when...
Trend	How does it change over time?	Line chart, area chart	Few discrete periods → column chart
Comparison	How do categories differ?	Bar chart, grouped bar	Many categories or long labels → dot plot
Ranking	What is highest or lowest?	Sorted horizontal bar	15–30 items → lollipop chart
Part-to-Whole	How does each part contribute?	Stacked bar, 100% stacked bar	2–4 parts, one moment → careful pie/donut
Distribution	How are values spread?	Histogram, box plot	Shape matters or n is small → violin / strip
Relationship	How are variables related?	Scatter plot, heatmap	A third variable matters → bubble chart
Geographic	Where does it happen?	Choropleth (rates), symbol map	<15 regions, location not the point → bar
Target / Deviation	How do we compare with plan?	Bar + target line, bullet chart	The gap is the message → variance chart

Chart selection follows the question - not habit or tool defaults.



Effective Data Visualization Principles

Choose honestly

- **Fit chart to question** - geometry follows the analytical question; styling cannot rescue the wrong chart type.
- **Title states the finding** - axes carry units, so no verbal explanation is needed.
- **Honest axes and scales** - zero baseline for bars; linear by default, log only for multiplicative data.
- **Right grain, right order** - state the aggregation; sort by value unless categories have a natural order.

Reduce noise

- **Color must encode meaning** - highlight, group or scale - otherwise use a single color.
- **Cut clutter, never 3D** - delete gridlines, shadows and duplicate legends; perspective distorts the data.
- **Emphasize one thing** - mute the rest - if everything is emphasized, nothing is.
- **Context and accessibility** - show a benchmark or target; color-blind-safe palettes, strong contrast, readable type.